■ 1.4.9 Working with Large Symbolic Expressions

When you do symbolic calculations, it is very easy to end up with extremely complicated expressions. Often, you will not even want to *see* the complete result of a computation.

If you end your input with a semicolon, *Mathematica* will do the computation you asked for, but will not display the result. You can nevertheless use `%` or `Out[n]` to refer to the result.

Even though you may not want to see the *whole* result from a computation, you often do need to see its basic form. You can use `Short` to display the *outline* of an expression, omitting some of the terms.

Ending your input with ; stops *Mathematica* from displaying the complicated result of the computation.

```
In[1]:= Expand[(x + 5 y + 10)^4] ;
```

You can still refer to the result as %. `//Short` displays a one-line outline of the result. The `<<n>>` stands for *n* terms that have been left out.

```
In[2]:= % //Short

                                             4
Out[2]//Short= 10000 + 4000 x + <<12>> + 625 y
```

This shows a three-line version of the expression. More parts are now visible.

```
In[3]:= Short[%, 3]
Out[3]//Short=

                          2       3    4
   10000 + 4000 x + 600 x  + 40 x  + x  + 20000 y +

                             3           3        4
    6000 x y + <<5>> + 5000 y  + 500 x y  + 625 y
```

This gives the total number of terms in the sum.

```
In[4]:= Length[%]
Out[4]= 15
```

command ;	execute *command*, but do not print the result
expr // `Short`	show a one-line outline form of *expr*
`Short[`*expr*, *n*`]`	show an *n*-line outline of *expr*

Some ways to shorten your output.